



# Second Welfare Theorem, an application

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# Negishi's approach

- Use the welfare theorems to identify WE.
- Analysis is in utility space, not commodity space.
- Useful with many goods and few agents.
- Proposed by Negishi (Negishi 1960).
- Employed by Arrow and Hahn (Arrow and Hahn 1971) and many others.
- Overview in Mascolell-et al, (Mas-Colell, Whinston, and Green 1995, "16.E, 17.Appendix A")

# Preliminaries

- Let  $\mathcal{E}$  be an Arrow-Debreu economy.
- $(\forall i) (\exists u_i, u_i : X_i \rightarrow \mathbb{R})$  such that

$$(\forall x_i, x'_i \in X_i) [x_i \succeq_i x'_i \iff u_i(x_i) \geq u_i(x'_i)].$$

- Let  $x = (x_1, \dots, x_i, \dots, x_I)$ .
- $u(x)$  denotes the vector  $(u_1(x_1), \dots, u_I(x_I))$ .
- Let  $u : \mathbb{R}_+^{LI} \rightarrow \mathbb{R}^I$  be the map  $x \mapsto (u_1(x_1), \dots, u_I(x_I))$ .

# Utility possibility set

**Definition 1** The utility possibility set is

$$U = \{u \in \mathbb{R}^I : \exists \text{ an allocation } x \text{ with } u \leq u(x)\}$$

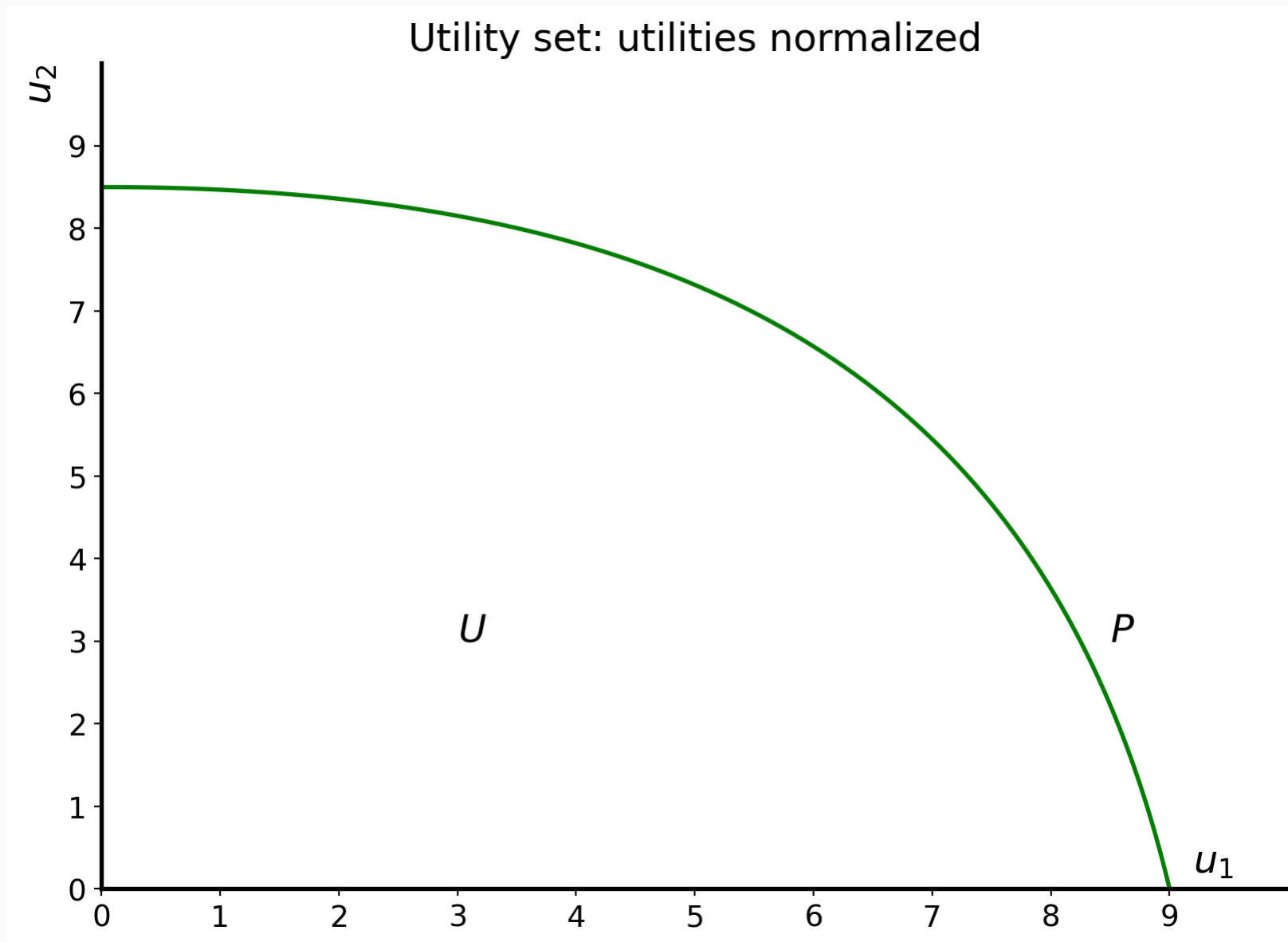
and its Pareto frontier is

$$P = \{u \in U : \nexists u' \in U, u' > u\}.$$

An allocation  $x^*$  is Pareto optimal if and only if  $u(x^*) \in P$ .

Similar definitions apply to economies with production.

# Utility possibility set



# Social welfare function

- Suppose society's values are summarized by a function

$$W(u) = W(u_1, \dots, u_I).$$

- Pick  $a \in \mathbb{R}^I$  with  $(\forall i) a_i \geq 0$ . Define

$$W(u) = \sum_{i=1}^I a_i u_i = a \cdot u.$$

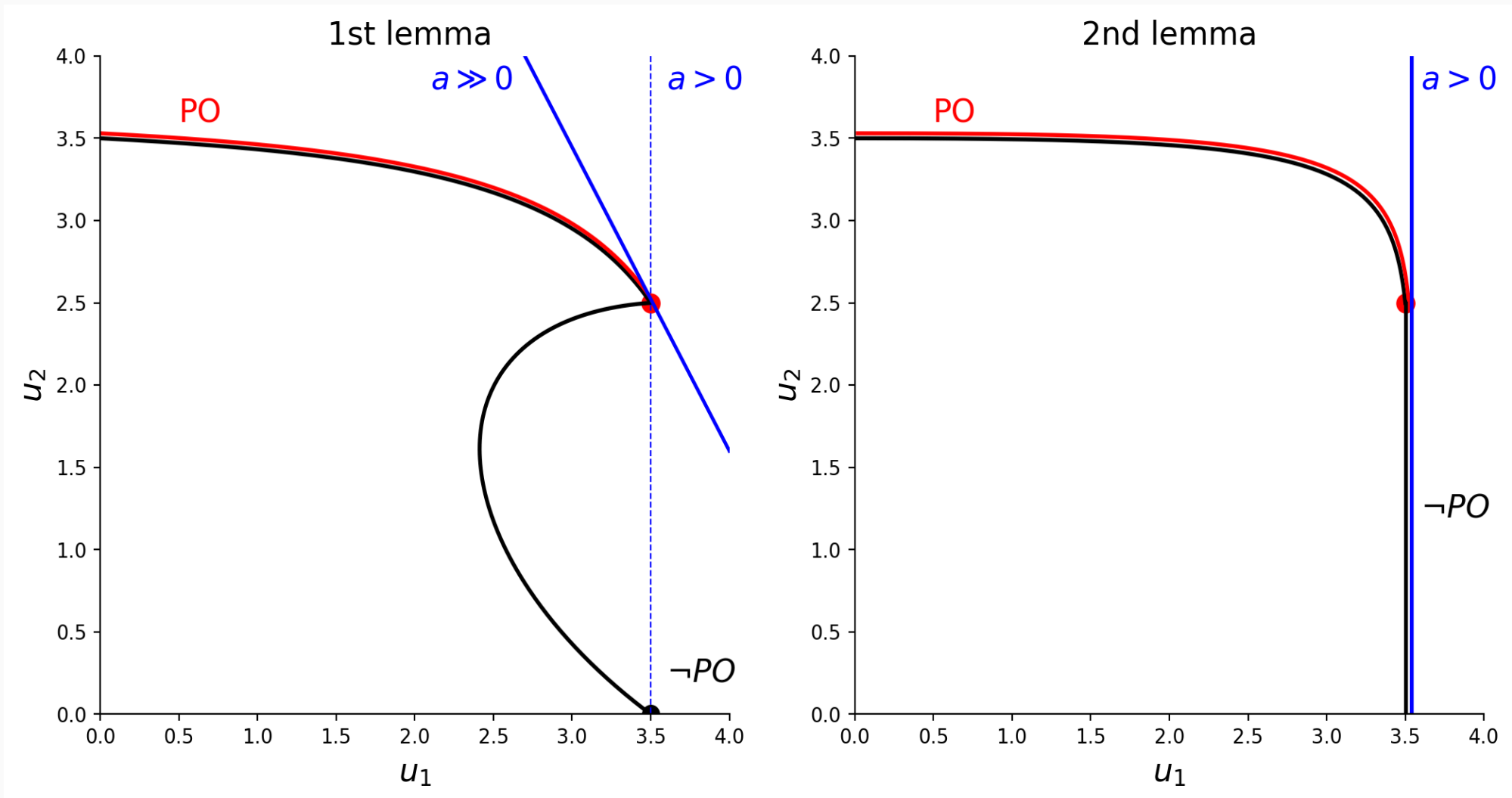
# Lemmas

**Lemma 1** If  $\exists a \in \mathbb{R}_{++}^I$  and  $u^* \in U$  such that  $(\forall u \in U) a \cdot u^* \geq a \cdot u$ , then  $u^* \in P$ .

**Lemma 2** If  $U$  is convex, then for any  $u^* \in P$ ,  $(\exists a \in \mathbb{R}_+^I) a \neq 0$  and  $(\forall u \in U) a \cdot u^* \geq a \cdot u$ .



# Lemma 1 and Lemma 2



# Stylized method

- Let  $\Delta = \{s \in \mathbb{R}_+^I : \sum_i s_i = 1\}$ .
- Fix  $a \in \Delta$ , pick  $u(a) \in U$  so that  $u(a) \in P$ .
- $u(a)$  is derived from an allocation  $(x(a), y(a))$ .
- The Weak Second Welfare Theorem implies
  - $\exists p(a), y_j(a)$  maximizes firm  $j$ 's profit and,
  - $x_i(a)$  is in  $i$ 's quasi demand.
- Identify transfer:  $(\forall i), t_i = p(a) \cdot [\omega_i - x_i(a)]$ .
- If  $(\exists a)$  such that  $(\forall i) t_i = 0$ . This identifies a WE.

Note Arrow's exceptional economy.

# References

- Arrow, Kenneth, and Frank H. Hahn. 1971. *General Competitive Analysis*. 1st ed. Vol. 12. North Holland.
- Mas-Colell, Andreu, Michael Whinston, and Jerry Green. 1995. *Microeconomic Theory*. New York, Oxford: Oxford University Press.
- Negishi, T. 1960. "Welfare Economics and the Existence of Equilibrium for a Competitive Economy." *Metroeconomica* 12: 92–97.